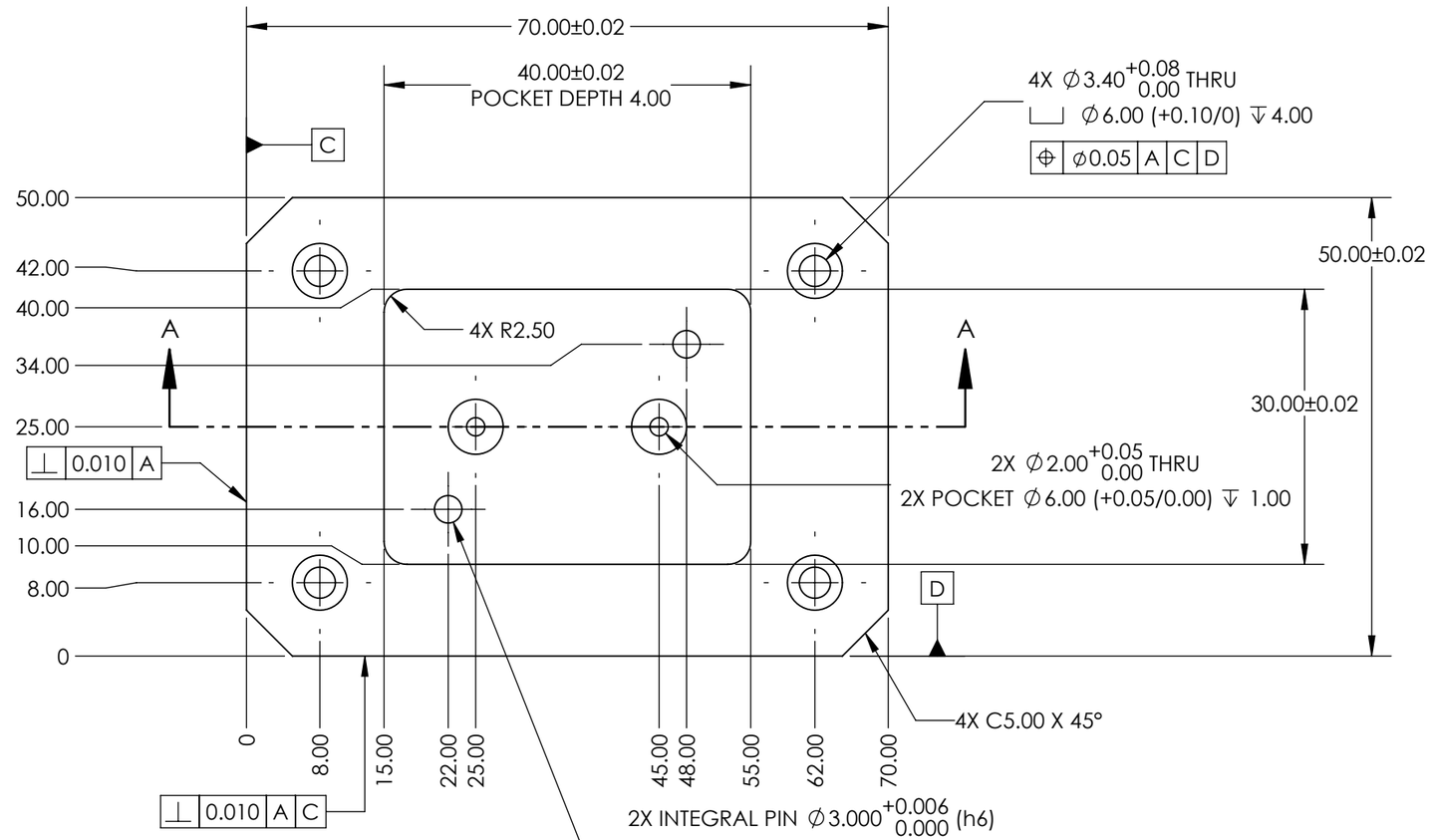
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-3C-PRECISION-JIG-1.3.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.02 mm, ANGULAR +/-0.1 DEG (HIGH-SPEED 5-AXIS CNC MILLING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

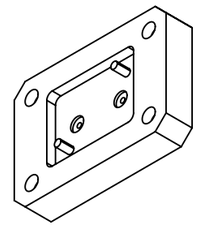
2. 3C ASSEMBLY & PART CONSOLIDATION ASSURANCE:
2.1 MATERIAL OPTION A: FORGED AEROSPACE ALUMINUM 7075-T6 (AMS 4045).
2.2 MATERIAL OPTION B: CARBON-FIBER REINFORCED ESD PEEK-CA30, SURFACE RESISTIVITY 10^6 - 10^9 OHM/SQ TO PREVENT ELECTROSTATIC DISCHARGE (ESD) DAMAGE TO DIE SURFACES.
2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING (SLM/SLS) OR POWDER SINTERING PERMITTED TO AVOID PARTICULATE CONTAMINATION.
2.4 PART CONSOLIDATION: 2X LOCATING PINS ARE MONOLITHICALLY INTEGRATED WITH BASE BODY TO ELIMINATE ASSEMBLY STACK-UP TOLERANCE (ZERO PIN PRESS-FIT ERROR).

3. KINEMATIC RIGIDITY & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (MACHINE MATING FACE): FLATNESS <= 0.005 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
3.2 SECONDARY DATUM [B] (PRODUCT SEATING FACE): PARALLEL TO [A] WITHIN 0.008 mm, FLATNESS <= 0.005 mm, Ra <= 0.4 MICRONS TO ENSURE ACCURATE ASSEMBLY ALIGNMENT.
3.3 2X INTEGRAL PINS (Ø3.000 +0.006/0.000 mm h6): CYLINDRICITY <= 0.002 mm, PERPENDICULAR TO [B] WITHIN 0.005 mm, TRUE POSITION WITHIN Ø0.008 mm RELATIVE TO [B], [C], [D].
3.4 4X CORNER MOUNTING HOLES: TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO [A], [C], [D]. MINIMUM WALL THICKNESS TO CORNER CHAMFER >= 4.50 mm.

4. CLEANING & SURFACE FINISH:
4.1 DEBURR 100% UNDER 30X MAGNIFICATION. ALL SHARP EDGES BROKEN C0.20 mm MAX.
4.2 SURFACE TREATMENT: FOR AL 7075: HARD ANODIZE TYPE III (MIL-A-8625, 20-25 µm THICK, PTFE SEALED FOR ULTRA-LOW WEAR); FOR PEEK: PRECISION ULTRASONIC DEGREASED.
4.3 CLEANROOM PACKAGING: ULTRASONICALLY CLEANED WITH DEIONIZED WATER, VACUUM SEALED IN DOUBLE PE BAGS FOR CLASS 1 / ISO 1 CLEANROOM COMPLIANCE.



SECTION A-A



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.02 mm, Angular ±0.1° (High-Speed 5-Axis CNC Milling)	FINISH: For Al 7075: Hard Anodize Type III (MIL-A-8625, 20-25 µm, Ultra-Clean PTFE Sealed) / For PEEK: Ultrasonic Degreased Class 1 Cleanroom Packaged		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE		TITLE: 3C HIGH-DENSITY ELECTRONIC PRECISION TOOLING JIG MOUNT	
DRAWN						
CHK'D						
APP'V'D						
MFG						
Q.A				MATERIAL: Option A: Forged Aerospace Aluminum 7075-T6 (AMS 4045) / Option B: ESD PEEK-CA30 (Carbon-Fiber Reinforced, Surface Resistivity 10^6 - 10^9 Ω/sq)	DWG NO. W3-3C-PRECISION-JIG-1.3	A3
				WEIGHT: ~100.8 g (Al 7075-T6) / ~50.2 g (ESD PEEK)	SCALE NTS	SHEET 1 OF 1